

TITLE

A PROJECT REPORT

Submitted by

STUDENT 1 NAME (ROLL NO)

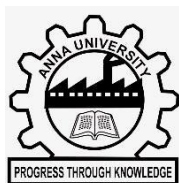
STUDENT 2 NAME (ROLL NO)

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of*

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND BUSINESS SYSTEMS



**RAJALAKSHMI ENGINEERING COLLEGE
ANNA UNIVERSITY: CHENNAI 600 025**

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RAJALAKSHMI ENGINEERING COLLEGE

CHENNAI

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Certified that this Project report titled “**TITLE**” is the bonafide work of “**STUDENT 1 NAME (ROLL NO), STUDENT 2 NAME (ROLL NO)**” who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

Waste management is a critical challenge faced by modern cities, often leading to environmental pollution and inefficient resource utilization. This project proposes a Smart Waste Management System that leverages Internet of Things (IoT) technology to optimize waste collection and disposal. The system utilizes smart bins equipped with ultrasonic sensors to monitor fill levels in real time. Data collected from these sensors is transmitted to a central server, where it is processed and analyzed using a web-based application.

The proposed model includes route optimization algorithms for waste collection vehicles, reducing fuel consumption and operational costs. A mobile application is also developed to provide updates to municipal authorities and allow citizens to report waste-related issues.

The initial implementation demonstrates accurate fill-level detection with an error margin of less than 5%, along with a 20% improvement in collection efficiency during simulations. The project aims to enhance environmental sustainability, reduce operational expenses, and improve public hygiene.

Keywords:

Note:

An abstract is a brief summary of your project that helps the reader understand:

- 1. Purpose – Why the project is being done (problem statement).**
- 2. Method – How you approached the problem (methods, tools, process).**
- 3. Outcome – What results or progress you have achieved.**
- 4. Significance – Why this work matters (impact or benefit).**

Length: 150–250 words

ACKNOWLEDGEMENT

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STUDENT 1 NAME

STUDENT 2 NAME

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CHAPTER-1

INTRODUCTION

1.1 GENERAL

Crime rates have increased tremendously. According to the National Crime Records Bureau, a total of 42,54,356 Indian Penal Code (IPC) crimes were registered in 2021. This is a 28% increase compared to previous year rates. The Norton Cyber Safety Insights Report released in April, 2021 revealed that there has been a distinct increase in the number of cyber attacks on individual and high maintenance systems. With current crime management softwares storing and hosting data in cloud platforms, the attack surface has expanded massively. These systems are thus striving to prevent the possibility of being prone to data breaches and cyber attacks in the process of digital adoption of crime management. With digital forensics and criminal analysis, there is a need for shifting from physical to digital evidence. This leads to an issue of massive data storage and retrieval. Crime management systems rely on the principle of immediate and quick data retrieval without tampering the authenticity of data. Outsourcing for encryption and firewalls might lead to data leak and breaches as there is no central administration of the system.

Yet another difficulty is the consolidation and reporting of complete crime patterns of the country. As of 2022, consolidation of these reports is done by the Ministry of Home Affairs by collecting incident data from each state. The NCRB collects this data manually and compiles it into a “Crime In India” report. This manual segregation, consolidation and reporting of crimes follows a “Principle Offense Rule” where only the most heinous and intense crimes is considered as a counting

unit in its data. This is a major concern as it reduces the accuracy of the reports for further analysis. Introducing machine learning techniques to consolidate every single crime reported by managing a central database architecture. Segregation of these crimes by storing them in buckets within the database can optimize the dataset fed in the machine learning algorithm. The identification of these issues in the existing architectures paved way for a research to create a solution that encrypts sensitive data and case documents with quick retrieval and manipulation of data by authenticated sources whilst maintaining the integrity of the system and its data.

1.2 OBJECTIVES

The aim of the project is to securely handle the digitized crime records that are on the police records removing the erroneous or tedious manual processes in maintaining crime records and classifying them providing an improved security system.

- To make sure the crime records and case documents are digitized.
- To ensure data integrity of the digital FIRs
- Avoid Data Tampering and unauthorized access to crime records
- One-stop solution for searching, updating, and securing records
- Securing the data using Encryption

1.3 EXISTING CRIME REPORTING SYSTEMS

Over the years the crime rate has increased tremendously, with case registrations increasing by 1.6% annually in India each year. During the pandemic, we needed a digital-based reporting system to register and process complaints. This is why these systems have grown so quickly. These reporting systems have been around for decades, but have been gradually enhanced with interactive features since 2017. Most government portals have many known bugs and shortcomings. To deal with

this government, there is a portal called NCIIPC's Responsible Vulnerability Disclosure Program (RVDP) where citizens can report bugs in the system. For cybercrime reporting, there is an SSL-encrypted Cybercrime.gov.in with a separate section for women/children crime complaint portals where administrators focus users on the backend complaints. There is also a tracking feature for police officers to update the status of these complaints. This portal is the same for all states, but if you want to report an unrecognized crime specifically in Tamil Nadu, view and file your complaint at eservices.tnpolice.gov.in and provide a copy of your First Information Report (FIR). Available. This system may also experience server downtime or it may take some time to load and save the details entered by the user, but hackers have destroyed so many advanced technologies that these Captchas do not work to create a secure system.

These systems mostly employ SSL/TLS encryption. Data sent over the internet or a computer network is protected by SSL (Secure Sockets Layer) encryption and TLS (Transport Layer Security), which is a more recent and secure successor. By doing this, data sent between two nodes—typically a user's web browser and a web/app server—is shielded from attackers (and Internet Service Providers) accessing or manipulating. The majority of website owners and operators are required to use SSL/TLS to secure the transmission of sensitive data, including passwords, payment information, and other private information. Therefore, creating an application that secures the police end would also automatically secure the user end.

1.3 LIMITATIONS IN EXISTING SYSTEM

- ❖ Lack of security enhancement to store sensitive data
- ❖ Requires High development cost and advanced system configurations to deploy
- ❖ Applicability to design digital witness for each design of physical network, reducing expandability

- ❖ Does not secure the Key while using Symmetric Key Encryption Algorithms
- ❖ Complicated Deployment and inefficiency while running in incompatible systems.

1.4 PROPOSED SYSTEM

Numerous records from various police stations are still kept in numerous folders and files, but when natural disasters strike, the information may be lost or become out-of-date. Include the possibility of easy removal and modification of these records, despite the fact that they are kept in a warehouse. Thus, they become more difficult and time-consuming to handle and maintain, or they can be added to an existing case to include newer crime records. Although the government has made significant efforts to raise awareness of the issue of digital data integrity, it is difficult to manage the security, storage, and practicality of all criminal records at once.

The aim is to create a secure system for smart analysis of crime reports through First Information Report (FIR) filed at stations and to build a digitized application for secure storage of records and to ease data segregation through type of crime classification

Cybersafe is a system that extracts information from digital FIR forms and identifies the formats for filing FIR reports and complaints. After the information is entered, a dropdown menu in the software gives users the opportunity to match the offense with the appropriate case parts. After the procedure is finished, the data are encrypted using the Hisea algorithm and saved in the cloud with various classifications.

Physical footprints and digital ones can both exist in crime records. The police will upload the criminal records as an image along with the case file to preserve the integrity of it. Even after a few days, if they decide to revisit or update them, they can open the app with their credentials and view the pool of records available; it

will then automatically point them in the direction of the pertinent criminal records. Links are used to store the digital footprint information rather than a complete description, and biometrics and multi-factor authentication are used to safeguard it.

The system workflow is as follows

- Authentication

The application uses keycloak authentication which is an open source Identity and Access Management tool where the users get authenticated through WebAuth Authentication along with Email verification and OpenID Access Grant with client authentication and authorization involved.

- Upload/View Crime Records

The application provides an access level control on which users get authorized to view the existing or newly filed complaints.

- Storage and Security

The crime records uploaded are secured and stored in an encrypted way.

- Encryption and Decryption

The case details are encrypted using the AES Algorithm and the Symmetric Key is protected using the HISEA Encryption Algorithm.

The features of the solution are

1. Encryption - Data are encrypted using Hisea and AES mechanism
2. Identity Management - Techniques like 2FA and user-group-admin classification
3. Log Analysis - Every login aspect is recorded so as to find malicious access

1.4.1 ADVANTAGES OF THE PROPOSED SYSTEM

- Enhanced Classification & Storage mechanism- Usual CMS has a master database where data is stored without primary classification. An innovative aspect is to classify data and store it into classified buckets enabling easier access and retrieval of data
- Frequent Crime analysis and visualization - Instead of a yearly audit and data

compilation, frequent crime analysis is carried out based on the data provided by the police in their FIRs such as higher crime locality, type of crime etc.

- **Most Secure algorithm for storage of case related documents** - AES, though is highly used has its own shortcomings. Innovation includes implementation of a more advanced encryption algorithm HISEA - Hybrid Cubes Encryption Security Enhancement Algorithm. Certified as the most secure encryption but lacks implementation.

CHAPTER - 2

LITERATURE SURVEY

2.2 LITERATURE REVIEW

The systematic literature survey is intended to study the existing solutions and proposals pertaining to the domain and purpose of the project. The literature review is carried out in the following ways: characterization of the systematic literature review as practiced in crime reporting systems and as modified for use. Then, referencing a systematic literature review pertaining to the intended technology or algorithm used, explaining the development of our proposed solution, definition of inclusion/exclusion criteria, characterization of the solution, and synthesis of the findings.

2.1 ONLINE CRIME REPORTING SYSTEM

Archana M et.al. [1] talks about an online crime management system that allows victims to file a case by visiting the website which has classifications of various sections and the user will be enabled to upload the evidence. There is an option of SOS button in the app which will help them to fetch their location and forward their complaints to the nearest police station. It briefly explains the concept of cookies and IP Addressing to identify location and enable blocking of unauthorized access. The work established a bridge to understand the working of a CMS System. The identified scope of improvement was the security enhancement required while storing sensitive data and evidence files.

Ganashree G et.al. [4] formulates a new method to secure a crime case summary and E-FIR in a police station using block chain technology. AES

Rijndael algorithm is adapted to make the data secure through encryption and also be able to decrypt the file using the cipher key with which encryption is done. The problem identified was the difficulty in crime case summary storage and access. Another issue was the high development cost and inadequate infrastructure owed to the introduction of block chain technology.

Archana Iyer et.al. [14] proposed an E-Police application built for Android Operating system in which it comprises a portal for police to register complaints and also keep track of them. The police would be provided with a FIR application and after filling it the complaint would be registered. The interface is created using the ADT kit in Eclipse while WAMP server is used for server facility.

K. Tabassum et.al. [15] proposes a system which is built for Riyadh citizens and policemen to register complaints and report them to the officials effectively through an online portal. The aim of the application is to increase the usability of online portals to register complaints and effectively manage them through interactive application between the user and the police. It is an effective application to keep track of the criminals and have a record of them so that it is easy to map complaints with the respective criminals.

2.2 RESEARCH PAPERS ON DIGITIZATION OF POLICE SYSTEMS

T .M. Zaharchuk et.al. [2] conducted a study and the aim of the report is formulating a general roadmap consisting of procedures and rules for using online portals to register crimes assisting the Canadian municipal forces to organize their records. It extensively elaborated on the Technological Alternatives to enhance existing systems and a new outlook on the creation and implementation of the Police Management System(PMS) from a grass root level. The work covered an interpretative analysis of the requirements for evaluating the information systems and its security of the Canadian Police Management Systems.

Supriya Pawar et.al. [12] conducts a study on real-world applications of the Police Management System(PMS) where they understand the internal difficulties and corruption involved in the process of case filing. This is a survey paper for understanding and understanding the needs of the police by studying and analyzing the active police users and their technical proficiency, issues and physical challenges in context to the law-and-order situation in India.

2.3 AUTHENTICATION AND AUTHORIZATION

Al Abdulwahid et.al. [19] Wrote a research work primarily based on reviewing existing authentication techniques and also the authentication mechanisms used in real-world applications with the aim to build state-of-the-art systems to satisfy the needs and loopholes in the system. It also covers in depth areas of authentication technologies and its tradeoff between high security and user satisfiability. The outcome of this research work is to secure the user's data with authentication and provide them an easy interface to interact with the system satisfying both security and interact ability.

G. Grieco et.al. [20] demonstrates the current standardization of using proprietary methods for authentication and authorization "As-a-Service" products and the requirement for the implementation of an advanced on-premise solution. The work offers a module for authentication and authorization that automatically protects information passing between modules for cyber security frameworks. The suggested remedy has been put into practice and evaluated in a genuine cyber-security architecture being developed for the H2020 GUARD project. The suggested module provides delegation of authentication and permission procedures among GUARD modules, according to results.

Sanjar Ibrokhimov et.al. [5] covers the digital authentication system which has the capability to make secure entry to different digital systems that are on the site.

The work also covers about different area where the authentication system plays an important role and the improvement of added layer of security i.e. Multi Factor authentication that takes an input of username and password along with special codes that the users would normally receive at the time of logging into the system. Evolution of these authentication systems is read based on the five aspects such as Security, Privacy and usability factors providing a path for future research on new authentication mechanisms.

2.4 ENCRYPTION AND AUTHORIZATION

Sapiee Janel et.al. [6] The study expands on the idea of employing a non-binary block cipher using any and all combinations of hybrid cube layers as the source for the encryption and decryption keys. Diffusion functions and the substitution box (SBOX) are added to the encryption to increase its security. The research's experimental findings suggested that the suggested generator matrices from hybrid cube layers are suitable as contenders for a key schedule method. This study made it easier to put this cutting-edge encryption technique into practice. Large key spaces will complicate and lengthen brute force attacks on the key spaces.

Tushar et.al. [7] covers different cryptographic algorithms and their broad classifications namely symmetric and asymmetric key cryptography. Although most of these are implemented in real world systems there are quite a few loopholes which can be exploited reducing data security. Thus the authors formulated a new encryption and decryption algorithm which uses 100 key add on and addition of binary converted elements as 2's complement which gives an added security especially in the field of cloud computing but the disadvantage is that the algorithm works well on paper but it is very less efficient compared to other algorithms like AES.

Hailan Pan et.al [8] conducted research which is primarily based on digital image

encryption. Digital image encryption technology is studied with dual logistic chaotic maps as a tool. The simulation experiments are carried out and the results are analyzed from the histogram, pixel correlation, information entropy, key space size and key sensitivity. The work proposes an enhanced security effect in the digital image encryption world.

Khalique zafar et.al. [9] Conducted a study on encryption algorithms. Encryption acts as an important process in keeping the data secure which is in transit through computer networks or at rest stored in different hard drives of the storage system. With the rapid progression in data exchange through digital media and other channels keeping the information secure and confidential is important adhering to the compliance with the legal rules and regulations. With the increase in evolution of new algorithms cryptography has become a vital part for making the information complex and indecipherable. This paper talks about the cost-performance algorithms like Advanced Encryption Algorithm (AES), Triple DES, RSA and their effectiveness in data security.

Table 1 comprises encryption techniques which can be used to convert confidential data which is in plaintext to a scrambled form so that no intruder can access it. Right access according to job roles is important in protecting the data getting leaked outside the network

Table 1 - Encryption and Authentication

Author Name	Methodology	Performance Measure	Limitations
Abdulwahid et.al [19]	Comparative Study of Conventional Authentication Techniques like PIN , Cognitive Knowledge Question , Patterns , Token Based Approach like OTP with Current Use of Authentication System on Various Devices along with Single Sign On (SSO) and Federated Identity Analysis	Parameters like Token Based, Secret of the key, the entry point and the time taken to again Authenticate the user are evaluated for each of the device with advantages and disadvantages	It's a accumulation of all the authentication mechanisms present and it's current usage qualitatively not in terms of Speed or Throughput or encryption crack time is mentioned on the work
Sanjar Ibrokhimov et.al [5]	Multi Factor Authentication using - Fingerprints and User-Specific Random Projection - Threshold Cryptography - MultiModal Biometrics for Cloud Computing - Cloud Based Authentication	Comparison of MFA algorithms on parameters like usage of algorithms in Smart Card, OTP Functionalities , Cryptographic Technique , Biometric based systems and Tokens	It generally covers the concepts of authentication usage in real-world applications present in gadgets and it covers only the authentication systems which is an extension of web based systems not native to the mobile.
Giovanni Grieco et.al [20]	Authentication and Authorization module for data flowing on cyber security framework models	Tested on real cybersecurity framework H2020 GUARD	Less flexible in controlling the access control policies
Sapiee Janel et.al [6]	Hybrid Cubes Encryption Algorithm with Substitution Box and Diffusion Functions	Brute Force Mechanism to decrypt the key generation by the mentioned algorithm	Less feasibility in terms of implementing the algorithm in Software Applications
Pan. H et.al [8]	Chaos Technology - Double Logistics Chaotic Mapping	Encryption Effect of the Algorithm is tested on 10 sample images irrespective of it's image size , pixel width and height	It is tested only on Image not for Video or audio or Text based documents
Khalique Zafar et.al [9]	Study on Algorithms like RSA , DES , 3DES and AES	Comparative Study on parameters - Encryption and Decryption against File Size	Analysis is done on very few parameters and no cumulative results on the graphs were done.

T .M. Zaharchuk et.al [2]	Web Services for Police Management System (PMS) and Management Information System (MIS) , Targeted Information ·Processing System (TIPS) - General Design and Overview of Architecture	Comparison of Different Police Management Systems implemented in Canadian jurisdictions	Operational Costs and Maintenance of MIS based system is expensive
Ganashree G et.al [4]	AES Rijndael algorithm	Parameters like Length of the cryptography key and it's size along with the count it takes to encrypt a text , Encryption and Decryption Speed , Throughput , Effectiveness and Flexibility or modification and Level of Security	High Development Cost and inadequate infrastructure for blockchain based system
Archana Iyer et.al [14]	ADT Kit of Eclipse and WAMP Server	Android Application where the police should be able to add details of the case with related documents and able to view or modify or create new case documents	It uses unstabilized server version for storage of data and no authentication mechanisms in place to validate the officials who is entering the details regarding the case
K. Tabassum et.al [15]	PHP and MySQL for building online application named E-Cops	User Interface Testing and User Acceptance Testing is done and the features of the application are compared against it and finalizing the result with the expected result computing the Boolean value Yes/No	It has a limited number of functionalities and the User Interface is made using custom CSS which can be improved with leverage of third-party extensions like Bootstrap , Material UI.

Formatting Guidelines for Project Report

- **Font Style: Times New Roman**
- **Title Font Size: 14 (Bold)**
- **Content Font Size: 12 (Regular)**
- **Line Spacing: 1.5**
- **Alignment: Justified for content, Center-aligned for main headings**
- **Chapter Titles: Use uppercase, bold, and font size 14**
- **Subheadings: Use bold, font size 12**

Ensure uniform formatting throughout the document for a professional appearance.